



**Working Draft v0.1**

**MEF W175**

# **Circuit Impairment and Maintenance Notification API and Developer Guide**

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**June 2025**

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# List of Contributing Members

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The following members of the MEF participated in the development of this document and have requested to be included in this list.

| Member |
|--------|
|        |
|        |
|        |

Table 1. Contributing Members

## 1. Abstract

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This standard is intended to assist implementation of the Circuit Impairment and Maintenance Notification functionality defined for the LSO Cantata and LSO Sonata Interface Reference Points (IRPs), for which requirements and use cases are defined in MEF 113 *Trouble Ticketing Requirements and Use Cases* [MEF 113] and MEF 173 *CIM Notification Service Business Requirements and Use Cases* [MEF 173]. This standard consists of this document and complementary API definitions for CIM and CIM Notification.

This standard normatively incorporates the following files by reference as if they were part of this document, from the GitHub repository

<https://github.com/MEF-GIT/MEF-LSO-Sonata-SDK>

- `productApi/cim/circuitImpairmentAndMaintenance.api.yaml`
- `productApi/cim/circuitImpairmentAndMaintenanceNotification.api.yaml`

<https://github.com/MEF-GIT/MEF-LSO-Cantata-SDK>

- `productApi/cim/circuitImpairmentAndMaintenance.api.yaml`
- `productApi/cim/circuitImpairmentAndMaintenanceNotification.api.yaml`

## 2. Terminology and Abbreviations

---

This section defines the terms used in this document. In many cases, the normative definitions of terms are found in other documents. In these cases, the third column is used to provide the reference that is controlling, in other MEF or external documents.

In addition, terms defined in the standards referenced below are included in this document by reference and are not repeated in the table below:

- MEF 55.1 [[MEF 55.1](#)]
- MEF 55.1.1 [[MEF 55.1.1](#)]
- MEF 113 [[MEF 113](#)]
- MEF 150 [[MEF 150](#)]
- MEF 173 [[MEF 173](#)]

| Term                                                    | Description                                                                                                                                                                                                                                                                                                                                                     | Reference                                                |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Application Program Interface (API)                     | In the context of LSO, API describes one of the Management Interface Reference Points based on the requirements specified in an Interface Profile, along with a data model, the protocol that defines operations on the data and the encoding format used to encode data according to the data model. In this document, API is used synonymously with REST API. | <a href="#">[MEF 55.1]</a>                               |
| Buyer                                                   | In the context of this document, denotes the organization or individual acting as the customer in a transaction over a Cantata (Customer <-> Service Provider) or Sonata (Service Provider <-> Partner) Interface.                                                                                                                                              | This document; adapted from <a href="#">[MEF 55.1.1]</a> |
| Circuit Impairment and Maintenance Notification Service | A standards based fault management and maintenance API service provided by a Seller to a Buyer that provides timely notifications of Failure Situations, Service Impairments and Planned Maintenance on the Seller's network.                                                                                                                                   | <a href="#">[MEF 173]</a>                                |
| Failure Situation                                       | The Seller detects an occurrence of a Situation on a Product provided to the Buyer, where the Product is non-operational (for example, down, routing failures, severe packet drops, etc.).                                                                                                                                                                      | <a href="#">[MEF 173]</a>                                |
| Incident                                                | An entry within a Seller's tracking system created by the context of this document, denotes a situation that is not part of normal operation Seller, which contains information about a Situation in the Seller's network that has a possible negative impact on the operability of the network on a Product for one or more Buyers                             | <a href="#">[MEF 113]</a>                                |
| Issue                                                   | In the context of this document, denotes a problem with a Product as experienced by the Buyer that is not part of normal operation.                                                                                                                                                                                                                             | <a href="#">[MEF 113]</a>                                |
| Notification                                            | A message sent from the Seller to the Buyer to inform about an event that has occurred in regard to a specific instance of a Ticket or an Incident                                                                                                                                                                                                              | <a href="#">[MEF 113]</a>                                |
| Planned Maintenance                                     | Any scheduled maintenance to be performed by the Seller that may impact a Product for the Buyer, which also includes the planned ending time and date.                                                                                                                                                                                                          | <a href="#">[MEF 173]</a>                                |

|                   |                                                                                                                                                                                                                                                                                                                                                 |                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Requesting Entity | The business organization that is acting on behalf of one or more Buyers. In the most common case, the Requesting Entity represents only one Buyer and these terms are then synonymous.                                                                                                                                                         | [MEF150]                                |
| Responding Entity | The business organization that is acting on behalf of one or more Sellers. In the most common case, the Responding Entity represents only one Seller and these terms are then synonymous.                                                                                                                                                       | [MEF150]                                |
| REST API          | Representational State Transfer. REST provides a set of architectural constraints that, when applied as a whole, emphasizes scalability of component interactions, generality of interfaces, independent deployment of components, and intermediary components to reduce interaction latency, enforce security, and encapsulate legacy systems. | [REST]                                  |
| Seller            | In the context of this document, denotes the organization acting as the supplier in a transaction over a Cantata (Customer <-> Service Provider) or Sonata (Service Provider <-> Partner) Interface.                                                                                                                                            | This document; adapted from [MEF55.1.1] |
| Situation         | In the context of this document, denotes a problem that is not part of normal operation in the Seller's network                                                                                                                                                                                                                                 | [MEF 113]                               |
| Ticket            | An entry within a Seller's tracking system created by the Buyer (or a third party on behalf of the Buyer), which contains information about an Issue impacting normal operation of a Product, along with support interventions made by technical support staff, or third parties                                                                | [MEF 113]                               |

**Table 2. Terminology**

| Term     | Description                         | Reference |
|----------|-------------------------------------|-----------|
| API      | Application Program Interface       | [MEF55.1] |
| CIM      | Circuit Impairment and Maintenance  | [MEF 173] |
| REST API | Representational State Transfer API | [REST]    |

**Table 3. Abbreviations**

### 3. Compliance Levels

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 (RFC 2119 [RFC2119], RFC 8174 [RFC8174]) when, and only when, they appear in all capitals, as shown here. All key words must be in bold text.

Items that are **REQUIRED** (contain the words **MUST** or **MUST NOT**) are labeled as [Rx] for required. Items that are **RECOMMENDED** (contain the words **SHOULD** or **SHOULD NOT**) are labeled as [Dx] for desirable. Items that are **OPTIONAL** (contain the words **MAY** or **OPTIONAL**) are labeled as [Ox] for optional.

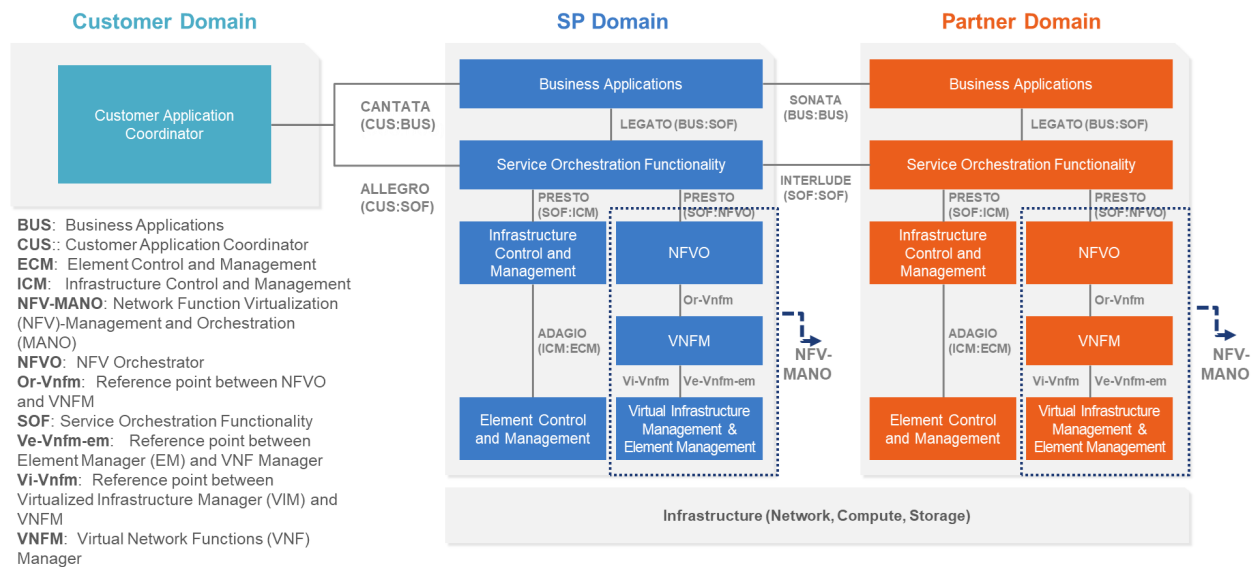
A paragraph preceded by [CRa]< specifies a conditional mandatory requirement that **MUST** be followed if the condition(s) following the "<" have been met. For example, "[CR1]<[D38]" indicates that Conditional Mandatory Requirement 1 must be followed if Desirable Requirement 38 has been met. A paragraph preceded by [CDb]< specifies a Conditional Desirable

Requirement that **SHOULD** be followed if the condition(s) following the "<" have been met. A paragraph preceded by **\*\*[COc]<\*\*** specifies a Conditional Optional Requirement that **MAY** be followed if the condition(s) following the "<" have been met.

## 4. Introduction

The CIM API allows the Buyer to list **CimIncidents**, get **CimIncident** details by **id** and register for receiving Notifications, and the Seller to send Notifications about circuit impairment incidents.

This standard specification document describes the Application Programming Interface (API) for CIM functionality of the LSO Cantata Interface Reference Point (IRP) and LSO Sonata IRP as defined in the *MEF 55.1 Lifecycle Service Orchestration (LSO): Reference Architecture and Framework* [MEF 55.1]. The LSO Reference Architecture is shown in Figure 1 with both IRPs highlighted.



**Figure 1. The LSO Reference Architecture**

Cantata and Sonata IRPs define pre-ordering and ordering functionalities that allow an automated exchange of information between business applications of the Buyer (Customer or Service Provider) and Seller (Service Provider or Partner) Domains. Those are:

- Product Catalog
- Address Validation
- Site Retrieval
- Product Offering Qualification
- Product Quote
- Product Inventory
- Product Ordering
- Trouble Ticketing
- Billing

The business requirements and use cases for CIM are defined in MEF 113 *Trouble Ticketing Requirements and Use Cases* [MEF 113] and MEF 173 *CIM Notification Service Business Requirements and Use Cases* [MEF 173]. MEF 113 defines use cases that cover Trouble Ticket, Incident, Appointment, and WorkOrder. MEF W175 defines additional requirements on top of those defined by MEF 113 for Incident. This document defines the CIM API as a separate, standalone single-purpose API and thus MEF W175 requirements may take precedence above ones defined in MEF 113. This document is structured as follows:

- **Chapter 4** provides an introduction to CIM and its description in a broader context of Cantata and Sonata and their corresponding SDKs.

- [Chapter 5](#) gives an overview of endpoints, resource model and design patterns.
- Use cases and flows are presented in [Chapter 6](#).
- And finally, [Chapter 7](#) complements previous sections with a detailed API description.

## 4.1. Conventions in the Document

- Code samples are formatted using code blocks. When notation `<< some text >>` is used in the payload sample it indicates that a comment is provided instead of an example value and it might not comply with the OpenAPI definition.
- Model definitions are formatted as in-line code (e.g. `CimIncident`).
- In UML diagrams the default cardinality of associations is `0..1`. Other cardinality markers are compliant with the UML standard.
- In the API details tables and UML diagrams required attributes are marked with a `*` next to their names.
- In UML sequence diagrams `{{variable}}` notation is used to indicate a variable to be substituted with a correct value.

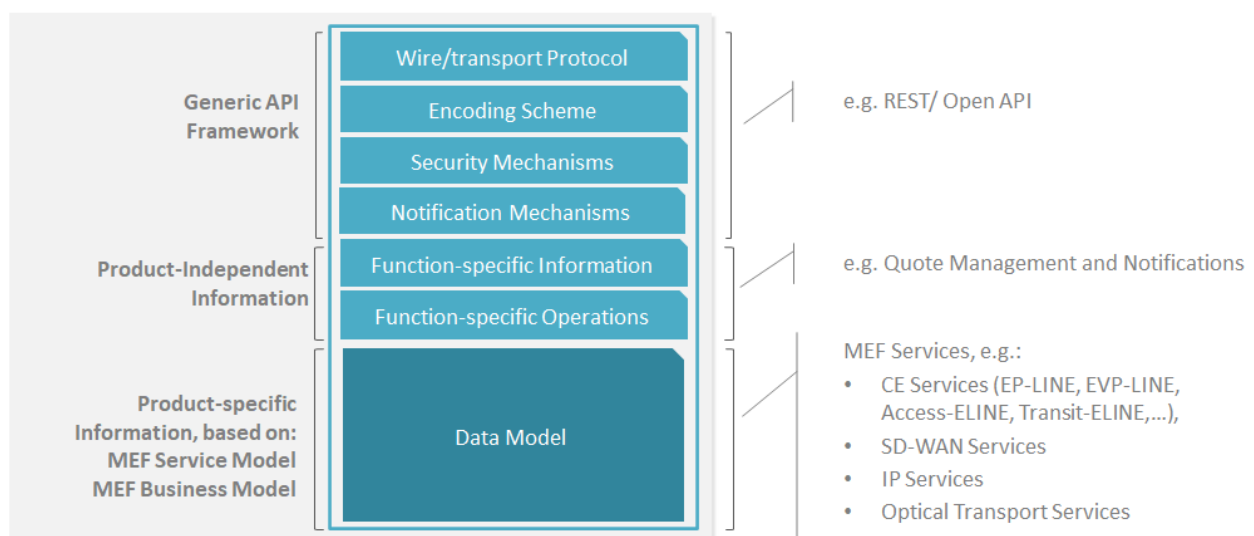
## 4.2. Relation to Other Documents

This API implements the CIM related requirements and use cases that are defined in MEF 113 [MEF 113] and MEF 173 *CIM Notification Service Business Requirements and Use Cases* [MEF 173]. This API definition builds on *TMF621 Trouble Ticket API REST Specification R19.0.1* [TMF621].

## 4.3. Approach

As presented in Figure 2, both Cantata and Sonata API frameworks consist of three structural components:

- Generic API framework
- Product-independent information (Function-specific information and Function-specific operations)
- Product-specific information (MEF product specification data model)



**Figure 2. Cantata and Sonata API framework**

The essential concept behind the framework is to decouple the common structure, information and operations from the specific product information content.

Firstly, the Generic API Framework defines a set of design rules and patterns that are applied across all Cantata or Sonata APIs.

Secondly, the product-independent information of the framework focuses on a model of a particular Cantata or Sonata functionality and is agnostic to any of the product specifications. Finally, the product-specific information part of the framework focuses on MEF product specifications that define business-relevant attributes and requirements for trading MEF subscriber and MEF operator services.

CIM API is product-agnostic in its nature and is not intended to carry any product-specific payloads. It only references products from the inventory by **id**. It operates using the Generic API Framework and the Function-specific Information and Operations.

#### 4.4. High-Level Flow

CIM is part of a broader Cantata and Sonata End-to-End flow. Figure 3 below shows a high-level diagram to get a good understanding of the whole process and CIM's position within it.



**Figure 3. Cantata and Sonata End-to-End Function Flow**

- Address Validation:
  - Allows the Buyer to retrieve address information from the Seller, including exact formats, for addresses known to the Seller.
- Site Retrieval:
  - Allows the Buyer to retrieve Geographic Site information including exact formats for Geographic Sites known to the Seller.
- Product Offering Qualification (POQ):
  - Allows the Buyer to check whether the Seller can deliver a product or set of products from among their product offerings at the geographic address or a Geographic Site specified by the Buyer; or modify a previously purchased product.
- Quote:
  - Allows the Buyer to submit a request to find out how much the installation of an instance of a Product Offering, an update to an existing Product, or a disconnect of an existing Product will cost.
- Product Order:
  - Allows the Buyer to request the Seller to initiate and complete the fulfillment process of an installation of a Product Offering, an update to an existing Product, or a disconnect of an existing Product at the address defined by the Buyer.
- Product Inventory:
  - Allows the Buyer to retrieve the information about the existing Product instances from Seller's Product Inventory.
- Trouble Ticketing:

- Allows the Buyer to create, retrieve, and update Trouble Tickets as well as receive notifications about Incidents' and Trouble Tickets' updates. This allows managing issues and situations for a Product provided by the Seller.
- Circuit Impairment and Maintenance:
  - Allows the Buyer to list CIM Incidents, get CIM Incidents details and register for receiveing Notifications, and the Seller to send Notifications about circuit impairment incidents

## 5. API Description

This section presents the API structure and design patterns. It starts with the high-level use cases diagram. Then it describes the REST endpoints with use case mapping. Next, it gives an overview of the API resource model.

### 5.1. High-level use cases

Figure 4 presents a high-level use case diagram as specified and numbered in MEF 173 [MEF 173] in section 7.2. Use cases are described extensively in [chapter 6](#).

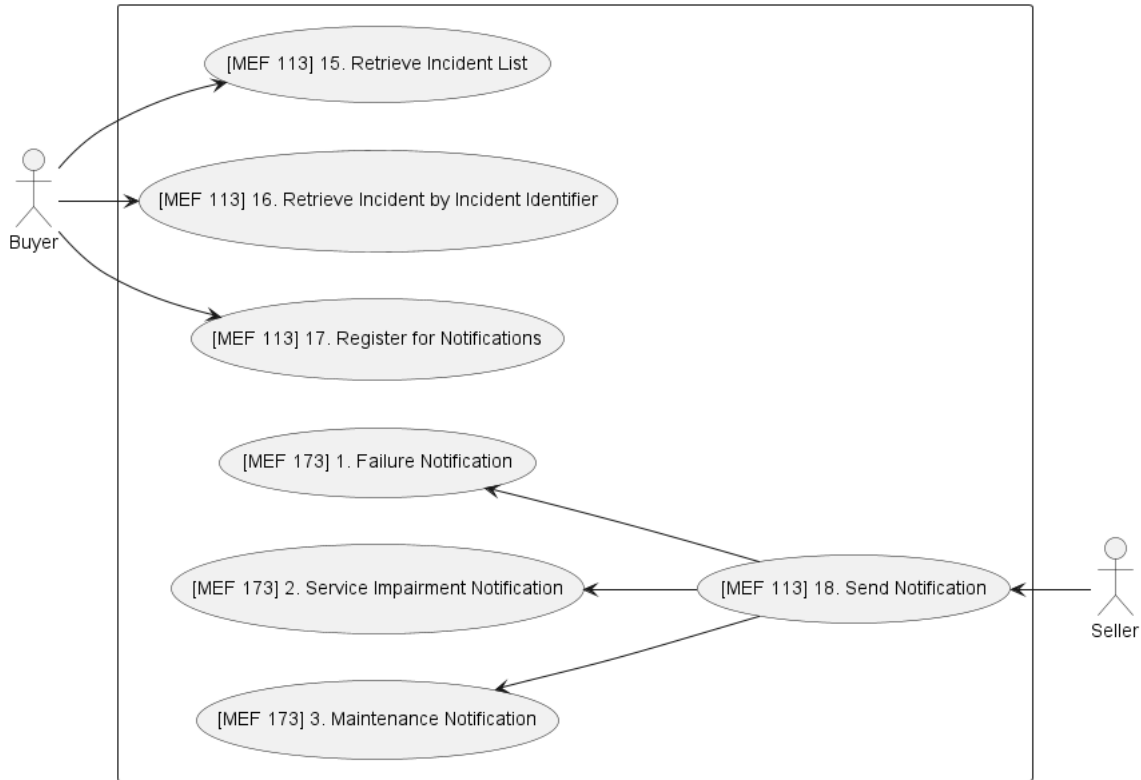


Figure 4. Use cases

### 5.2. API Endpoint and Operation Description

#### 5.2.1. Seller side API Endpoints

**Base URL for Cantata:**

`https://{serverBase}:{port}/{seller_prefix}/mefApi/sonata/cimIncident/v1/`

**Base URL for Sonata:**

`https://{serverBase}:{port}/{seller_prefix}/mefApi/cantata/cimIncident/v1/`

The following API endpoints are implemented by the Seller and allow the Buyer to list **CimIncidents**, get **CimIncident** details by **id** and register for receiving Notifications. The endpoints and corresponding data model are defined in:

`productApi\cim\circuitImpairmentAndMaintenance.api.yaml`.

| API endpoint            | Description                                                                                                                                                                 | MEF 113 / MEF 173 Use Case mapping                      |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| GET /cimIncident        | The Buyer requests a list of <b>CimIncidents</b> from the Seller based on a set of specified filter criteria. The Seller returns a summarized list of <b>CimIncidents</b> . | MEF 113 UC 15: Retrieve Incident List                   |
| GET /cimIncident/{{id}} | The Buyer requests detailed information about a single <b>CimIncident</b> based on a <b>CimIncident.id</b> .                                                                | MEF 113 UC 16: Retrieve Incident by Incident Identifier |
| POST /hub               | The Buyer requests to subscribe to notifications.                                                                                                                           | MEF 113 UC 17: Register for Event Notifications         |
| GET /hub/{{id}}         | A request initiated by the Buyer to retrieve the details of the notification subscription.                                                                                  | MEF 113 UC 17: Register for Event Notifications         |
| DELETE /hub/{{id}}      | A request initiated by the Buyer to instruct the Seller to stop sending notifications.                                                                                      | MEF 113 UC 17: Register for Event Notifications         |

**Table 4. Seller side mandatory API endpoints**

[R1] The Seller implementation **MUST** support API endpoints listed in Table 4. [MEF173 R19], [MEF173 R20]

### 5.2.2. Buyer side API Endpoints

**Base URL for Cantata:**

https://{{serverBase}}:{{port}}  
 {{?/buyer\_prefix}}/mefApi/sonata/cimIncidentNotification/v1/

**Base URL for Sonata:**

https://{{serverBase}}:{{port}}  
 {{?/buyer\_prefix}}/mefApi/sonata/cimIncidentNotification/v1/

The following API Endpoints are used by the Seller to post notifications to registered listeners. The endpoints and corresponding data model are defined in

productApi\cim\circuitImpairmentAndMaintenanceNotification.api.yaml

| API Endpoint | Description | MEF 113 Use Case Mapping |
|--------------|-------------|--------------------------|
|--------------|-------------|--------------------------|

| API Endpoint                                         | Description                                                                                         | MEF 113<br>Use Case<br>Mapping                                                                                                                                                |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POST /listener/CimIncidentCreateEvent                | A request initiated by the Seller to notify the Buyer on <b>CimIncident</b> creation                | MEF 113<br>UC 18: Send Notification<br>MEF 173<br>UC 1: Failure Notification<br>MEF 173<br>UC 2: Service Impairment Notification<br>MEF 173<br>UC 3: Maintenance Notification |
| POST /listener/CimIncidentAttributeValueChangedEvent | A request initiated by the Seller to notify the Buyer on <b>CimIncident</b> attribute value change. | MEF 113<br>UC 18: Send Notification<br>MEF 173<br>UC 1: Failure Notification<br>MEF 173<br>UC 2: Service Impairment Notification<br>MEF 173<br>UC 3: Maintenance Notification |
| POST /listener/CimIncidentStatusChangeEvent          | A request initiated by the Seller to notify the Buyer on <b>CimIncident.status</b> change.          | MEF 113<br>UC 18: Send Notification<br>MEF 173<br>UC 1: Failure Notification<br>MEF 173<br>UC 2: Service Impairment Notification<br>MEF 173<br>UC 3: Maintenance Notification |

**Table 5. Buyer side optional API endpoints**

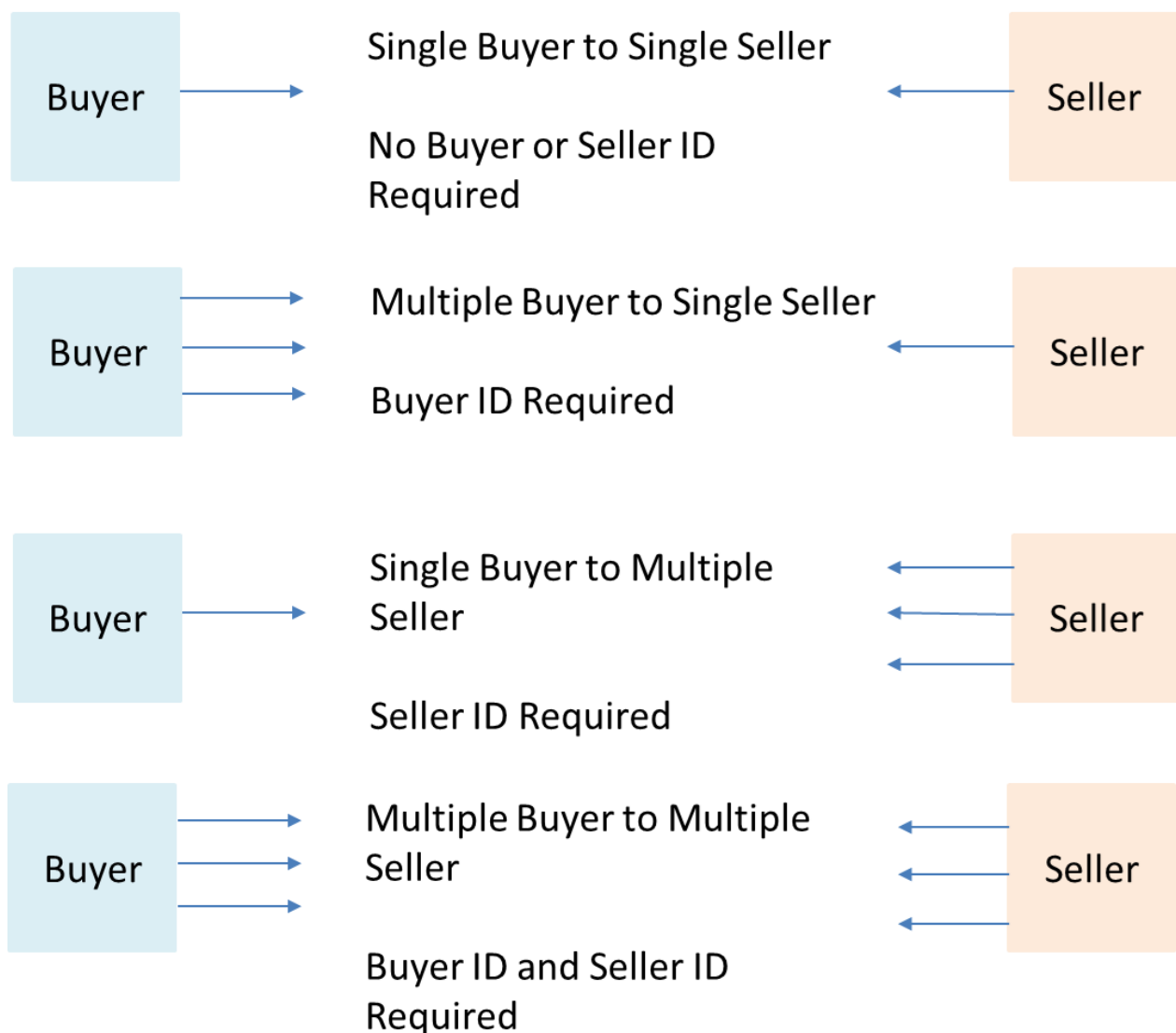
[D1] The Buyer implementation **SHOULD** support API endpoints listed in Table 5. [MEF113 D3]

[CR2]<([O1]) If any of endpoints listed in Table 5 is supported, then the Seller **MUST** support all endpoints listed in Table 5. [MEF113 [CR2]<[O2]]

### 5.3. Specifying the Buyer ID and the Seller ID

A business Entity willing to represent multiple Buyers or multiple Sellers must follow requirements of [MEF 150] chapter 8.8, which states:

For requests of all types, there is a business Entity that initiates an Operation (called a Requesting Entity) and a business Entity that is responding to this request (called the Responding Entity). In the simplest case, the Requesting Entity is the Buyer and the Responding Entity is the Seller. However, in some cases, the Requesting Entity may represent more than one Buyer and similarly, the Responding Entity may represent more than one Seller.



**Figure 5. Buyer ID and Seller ID Examples**

As shown in Figure 5, if a Requesting Entity representing a single Buyer is doing business with a Responding Entity representing a single Seller, Buyer and Seller IDs are not required to be passed between the two Entities. If a Requesting Entity representing more than one Buyer is doing business with a Responding Entity representing a single Seller, the

Buyer ID is required to be passed between the two Entities. If a Requesting Entity representing a single Buyer is doing business with a Responding Entity representing multiple Sellers, the Seller ID is required to be passed between the two Entities. If a Requesting Entity representing multiple Buyers is doing business with a Responding Entity representing multiple Sellers, both the Buyer ID and the Seller ID are required to be passed between the Entities.

While it is outside the scope of this specification, it is assumed that the Requesting Entity and the Responding Entity are aware of each other and can authenticate requests initiated by the other party. It is further assumed that the Buying Entity knows:

- the list of Buyers the Requesting Entity represents when interacting with this Responding Entity; and
- the list of Sellers that this Responding Entity represents to this Requesting Entity.

It is also assumed that the Responding Entity knows:

- the list of Sellers that this Responding Entity represents to this Requesting Entity and
- the list of Buyers the Requesting Entity represents when interacting with this Responding Entity.

In the API the **buyerId** and **sellerId** are represented as optional query parameters in each operation defined.

**[R2]** If the Requesting Entity has the authority to represent more than one Buyer the request **MUST** include **buyerId** that identifies the Buyer being represented [MEF150 R62]

**[R3]** If the Responding Entity represents more than one Seller to this Buyer the request **MUST** include **sellerId** that identifies the Seller with whom this request is associated [MEF150 R63]

## 5.4. Model Structural Validation

The structure of the HTTP payloads is defined using OpenAPI version 3.0.

**[R4]** Implementations **MUST** use payloads that conform to these definitions.

**[R5]** The Buyer and the Seller **MUST NOT** use any operation, entity or attribute that is not explicitly defined or allowed by this standard.

## 5.5. Security Considerations

There must be an authentication mechanism whereby a Seller can be assured who a Buyer is and vice-versa. There must also be authorization mechanisms in place to control what a particular Buyer or Seller is allowed to do and what information may be obtained. However, the definition of the exact security mechanism and configuration is outside the scope of this document. Security considerations are standardized by *LSO API Security Profile* [MEF 128.1].

## 6. API Interactions and Flows

---

This section provides a detailed insight into the API functionality and in the following subchapters describes the API usage flow and examples for each of the use cases.

### 6.1. Use Case 15: Retrieve Incident List

[O1] The Buyer **MAY** retrieve a list of Incidents by using a `GET /cimIncident` operation with desired filtering criteria. The attributes that are available to be used are: [MEF113 O20]

- `priority`
- `severity`
- `impact`
- `incidentType`
- `status`
- `relatedEntityId`
- `relatedEntityType`
- `creationDate.gt`
- `creationDate.lt`
- `situationStartDate.gt`
- `situationStartDate.lt`
- `expectedClosedDate.gt`
- `expectedClosedDate.lt`
- `closedDate.gt`
- `closedDate.lt`

The Buyer may also ask for pagination of the response when the number of results is too big. The following query attributes related to pagination can be provided:

- `limit` - number of expected list items
- `offset` - offset of the first element in the result list.

```
https://serverRoot/mefApi/sonata/cimIncidentNotification/v1/cimIncident?
status=inProgress&priority=critical&limit=10&offset=0
```

The example above shows a Buyer's request to get all `CimIncidents` that are in the `inProgress` status and with `critical` priority. Additionally, the Buyer asks only for a first (`offset=0`) pack of 10 results (`limit=10`) to be returned. The correct response (HTTP code `200`) in the response body contains a list of `CimIncident_Find` objects matching the criteria. To get more details (e.g. the item level information), the Buyer has to query a specific `CimIncident` by `id`.

The Seller returns a list of elements that comply with the requested `limit`. If the requested `limit` is higher than the supported list size then the smaller list of results is returned. In that case, the size of the result is returned in the header attribute `X-Result-Count`. The Seller can indicate that there are additional results available using:

- `X-Total-Count` header attribute with the total number of available results
- `X-Pagination-Throttled` header set to `true`

[D2] The Seller **SHOULD** support the pagination mechanism.

[CR3]<[D2] Seller **MUST** use either `X-Total-Count` or `X-Pagination-Throttled` to indicate that the page was truncated and additional results are available.

[R6] The Seller **MUST** put the following attributes into the **CimIncident** object in the response: [MEF113 R126]:

- @type
- creationDate
- description
- id
- impact
- incidentType
- priority
- relatedContactInformation - items with role equal to incidentContact
- relatedEntity
- severity
- situationStartDate
- status
- statusChange

[R7] In case no items matching the criteria are found, the Seller **MUST** return a valid response with an empty list. [MEF113 R127]

[D3] If the Buyer requests a list of new CIM Incidents, the Buyer **SHOULD** include a query param of **creationDate.gt** with a DateTime after the most recent Retrieve CIM Incident List poll request. [MEF173 D4]

```
https://serverRoot/mefApi/sonata/cimIncidentNotification/v1/cimIncident?creationDate.gt=lastPollRequestDateTime
```

[D4] If the Buyer requests a list of open CIM Incidents, the Buyer **SHOULD** include a query param of **status** with **created** and **inProgress** values. [MEF173 D5]

```
https://serverRoot/mefApi/sonata/cimIncidentNotification/v1/cimIncident?status=created&status=inProgress
```

[D5] If the Buyer requests a list of recently closed CIM Incidents, the Buyer **SHOULD** include a query param of **status** with **closed** value and **closedDate.gt** with a DateTime starting after the most recent Retrieve CIM Incident List poll request. [MEF173 D6]

```
https://serverRoot/mefApi/sonata/cimIncidentNotification/v1/cimIncident?status=closed&closedDate.gt=lastPollRequestDateTime
```

## 6.2. Use Case 16: Retrieve CIM Incident by Identifier

The Buyer can get detailed information about the Incident from the Seller by using a **GET /cimIncident/{id}** operation.

[R8] In case **id** does not allow to find an **CimIncident** instance, an error response **Error404** **MUST** be returned. [MEF113 R129]

[R9] The Seller **MUST** put the following attributes into the **CimIncident** object in the response: [MEF113 R131]

- @type
- creationDate
- description

- `id`
- `impact`
- `incidentType`
- `priority`
- `relatedContactInformation` - items with `role` equal to `incidentContact`
- `relatedEntity`
- `severity`
- `situationStartDate`
- `status`
- `statusChange`

[R10] The `statusChange` **MUST** include a full object's state history including the initial state.

[R11] The Seller **MUST** provide all remaining optional attributes if they are set. [MEF113 R132]

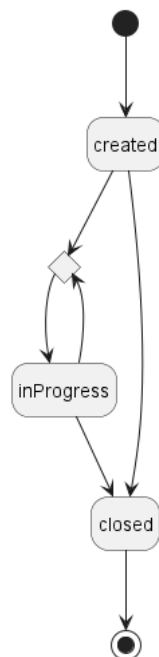
[R12] The Seller's **MUST** provide the `CimIncident.closedDate` if the `status` is `closed`. [MEF113 R133], [MEF173 R18]

Table 6 presents the mapping between the API `status` names and the MEF 113 naming, together with their description.

| status                  | MEF 113 name | Description                                                                                 |
|-------------------------|--------------|---------------------------------------------------------------------------------------------|
| <code>created</code>    | CREATED      | A new Incident has been created and allocated a unique <code>id</code> .                    |
| <code>inProgress</code> | IN_PROGRESS  | The Incident is in the process of being handled by the Seller.                              |
| <code>closed</code>     | CLOSED       | The Situation described in the Incident was closed by the Seller. This is a terminal state. |

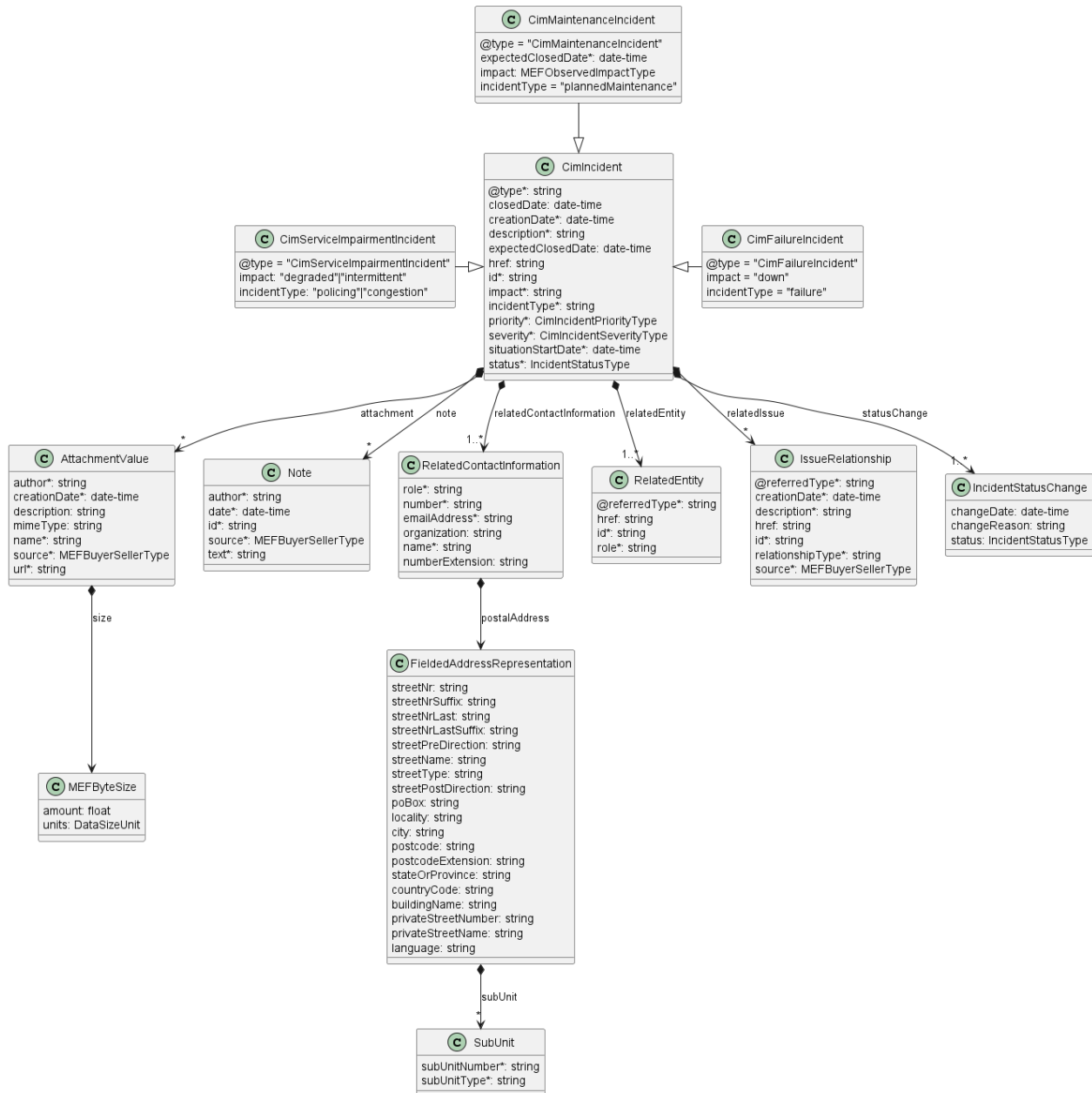
**Table 6. Incident states**

Figure 6 presents the Incident state machine:



**Figure 6. Incident State Machine**

[R13] The Seller **MUST** support all CIM Incident statuses and their associated transitions as described in Figure 6 and Table 6. [MEF113 R167]



**Figure 7. Use Case 16: CIM Incident Model**

```

{
  "@type": "CimFailureIncident",
  "id": "00001111-4321-6666-7777-000000003333",
  "href": "{{baseUrl}}/cimIncident/00001111-4321-6666-7777-000000003333",
  "attachment": [
    {
      "author": "Luke Example",
      "creationDate": "2025-06-12T23:09:44.814Z",
      "description": "Print screen from the assurance system",
      "mimeType": "image/jpeg",
      "name": "Alarm",
      "url": "https://seller.mef.com/documents/00000000-5555-4444-3333-222211110000",
      "size": {
        "amount": 2.6,
        "units": "MBYTES"
      },
      "source": "seller"
    }
  ],
  "creationDate": "2025-06-12T23:09:44.814Z",
  "description": "Hardware failure",
  "expectedClosedDate": "2025-06-13T23:09:44.814Z",
  "impact": "down",
  "incidentType": "failure",
  "situationStartDate": "2025-06-12T23:01:44.815Z",

```

```

"priority": "critical",
"relatedContactInformation": [
  {
    "emailAddress": "Incident.Contact@seller.mef.com",
    "name": "Incident Contact",
    "number": "+98-765-432-10",
    "organization": "Seller Example Co.",
    "role": "incidentContact"
  }
],
"relatedEntity": [
  {
    "id": "01494079-6c79-4a25-83f7-48284196d44d",
    "href": "https://mef.seller.com/mefApi/sonata/productInventory/v8/product/01494079-6c79-4a25-83f7-48284196d44d",
    "role": "Affected Product",
    "@referredType": "Product"
  }
],
"severity": "extensive",
"status": "created",
"statusChange": [
  {
    "changeDate": "2025-06-12T23:09:44.814Z",
    "status": "created"
  }
]
}

```

### 6.3. Use case 17: Register for Event Notifications

[R14] The Seller **MUST** support all types of events: [MEF173 R21]

- `cimIncidentAttributeValueChangeEvent`
- `cimIncidentCreateEvent`
- `cimIncidentStatusChangeEvent`

[D6] The Buyer **SHOULD** register for all types of events. [MEF173 D3], [MEF173 D7]

To register for notifications the Buyer uses the `registerListener` operation from the API: `POST /hub`. The request model contains only 2 attributes:

- `callback` - mandatory, to provide the callback address the events will be notified to,
- `query` - optional, to provide the required types of event.

The usage of a combination of these attributes and the `DELETE /hub/{id}` action fulfills the [MEF113 R137], [MEF113 R138], [MEF113 R139], [MEF173 R13], [MEF173 R14] requirements.

By using a simple request:

```

{
  "callback": "https://buyer.mef.com/listenerEndpoint"
}

```

the Buyer subscribes for notification of all types of events.

Note that the subscription can be differentiated only per event type (create, statusChange, attribute value change) but not per the CIM Incident type (failure, planned maintenance, impairment).

If the Buyer wishes to receive only notification of a certain type, a `query` must be added:

```
{
  "callback": "https://buyer.mef.com/listenerEndpoint",
  "query": "eventType=cimIncidentStatusChangeEvent"
}
```

If the Buyer wishes to subscribe to 2 different types of events, there are 2 possible syntax variants [TMF630]:

```
eventType=cimIncidentCreateEvent,cimIncidentStatusChangeEvent
```

or

```
eventType=cimIncidentCreateEvent&eventType=cimIncidentStatusChangeEvent
```

The **query** formatting complies to RCF3986 RFC3986. According to it, every attribute defined in the Event model (from notification API) can be used in the **query**. However, this standard requires only **eventType** attribute to be supported.

**[R15]** **eventType** is the only attribute that the Seller **MUST** support in the query.

The Seller responds to the subscription request by adding the **id** of the subscription to the message that must be further used for unsubscribing.

```
{
  "id": "00000000-0000-0000-0000-000000000678",
  "callback": "https://buyer.mef.com/listenerEndpoint",
  "query": "eventType=cimIncidentStatusChangeEvent"
}
```

Example of a final address that the Notifications will be sent to (for Sonata, **cimIncidentStatusChangeEvent**):

- <https://buyer.mef.com/listenerEndpoint/mefApi/sonata/circuitImpairmentAndMaintenanceNotification/v1/listener/cimIncidentStatusChangeEvent>

To stop receiving events, the Buyer has to use the **unregisterListener** operation from the **DELETE /hub/{id}** endpoint. The **id** is the identifier received from the Seller during the listener registration.

## 6.4. Use case 18: Send Event Notification

Notifications are used to asynchronously inform the Buyer about the respective objects and attributes changes.

**[R16]** The Seller **MUST** send Notifications of **eventTypes** to Buyers who have registered for them. [MEF113 R141], [MEF173 R15]

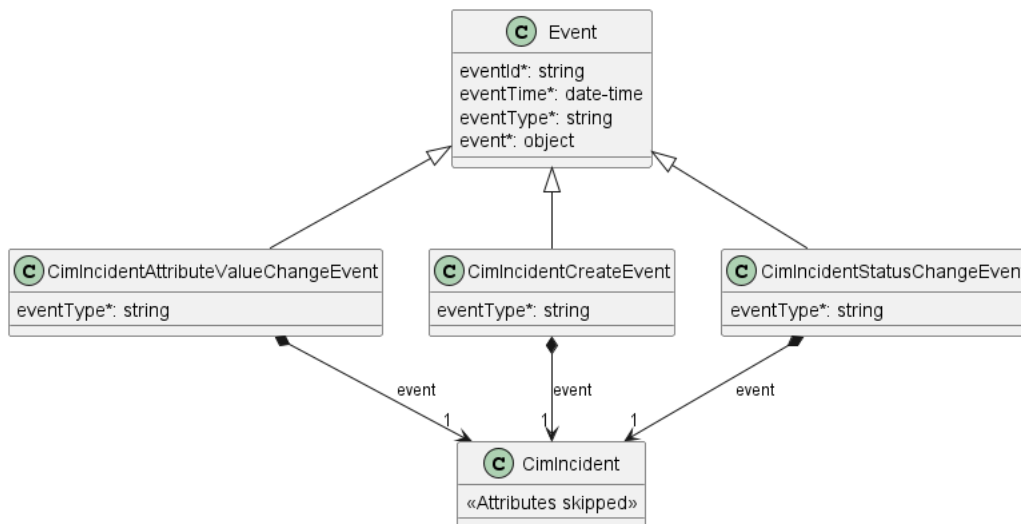
**[R17]** The Seller **MUST NOT** send Notifications for **eventTypes** to Buyers who have not registered for them. [MEF113 R140], [MEF173 R15]

**[R18]** The Seller **MUST** provide the following attributes of **CimIncident**: [MEF113 R131], [MEF173 R116]

- @type
- creationDate
- description
- id
- impact
- incidentType
- priority
- relatedContactInformation - items with role equal to incidentContact
- relatedEntity
- severity
- situationStartDate
- status
- statusChange

[R19] The Seller **MUST** provide all the remaining attributes of **CimIncident** instance if they are set. [MEF173 R17]

The Figure 8 shows entities involved in the Notification use cases. Every Event type has its own dedicated schema. Every type of Event carries the full representation of CimIncident as the event Payload. Details of **CimIncident** are skipped as they are same as on Figure 7.



**Figure 8. Notification Data Model**

The following snippet presents an example of **cimIncidentStatusChangeEvent**

```

{
  "eventId": "event-001",
  "eventType": "cimIncidentStatusChangeEvent",
  "eventTime": "2025-06-13T11:00:00Z",
  "event": {
    "@type": "CimServiceImpairmentIncident",
    "id": "INC-67890",
    "creationDate": "2025-06-13T09:00:00Z",
    "description": "Service is degraded due to congestion.",
    "impact": "degraded",
    "incidentType": "congestion",
    "priority": "medium",
    "relatedContactInformation": [
      {
        "emailAddress": "Incident.Contact@seller.mef.com",
        "name": "Incident Contact",
        "number": "+98-765-432-10",
        "organization": "Seller Example Co.",
        "role": "incidentContact"
      }
    ]
  }
},

```

```

    "relatedEntity": [
      {
        "@referredType": "Product",
        "id": "PROD-002",
        "href": "https://mef.seller.com/mefApi/sonata/productInventory/v8/product/PROD-002"
      }
    ],
    "severity": "moderate",
    "situationStartDate": "2025-06-13T10:45:00Z",
    "status": "InProgress",
    "statusChange": [
      {
        "changeDate": "2025-06-13T09:00:00Z",
        "changeReason": "Incident created",
        "status": "created"
      },
      {
        "changeDate": "2025-06-13T11:00:00Z",
        "changeReason": "Incident under investigation",
        "status": "InProgress"
      }
    ]
  }
}

```

The table below presents the mapping between the API Notification types' names and the ones in MEF 113 together with event descriptions. The inconsistencies are caused by API naming convention and using the TMF event types as the base for this API.

| API name                                          | MEF 113 name          | Description                                              |
|---------------------------------------------------|-----------------------|----------------------------------------------------------|
| <code>cimIncidentCreateEvent</code>               | INCIDENT_CREATE       | A new CIM Incident was created by the Seller.            |
| <code>cimIncidentAttributeValueChangeEvent</code> | INCIDENT_UPDATE       | An open CIM Incident was updated by the Seller.          |
| <code>cimIncidentStatusChangeEvent</code>         | INCIDENT_STATE_CHANGE | An CIM Incident <b>status</b> was changed by the Seller. |

**Table 7. Notification types mapping**

[R20] The Seller **MUST** send a `cimIncidentCreateEvent` whenever a new `CimIncident` has been created. [MEF113 R168]

[R21] The Seller **MUST** send a `cimIncidentAttributeValueChangeEvent` whenever the Seller updates any of the `CimIncident` attributes (excluding `status`) [MEF113 R169]

[R22] The Seller **MUST** send a `cimIncidentStatusChangeEvent` whenever a `CimIncident.status` change occurs. [MEF113 R170]

[R23] When the `CimIncident.status` moves to `InProgress`, the Seller **MUST** set the `expectedClosedDate`. [MEF113 R171]

[R24] The Seller **MUST NOT** send a Notification to a Buyer for a `CimIncident` impacting a Product that the Seller has not activated on behalf of the Buyer. [MEF113 R172]

## 6.4.1 Use case 1: Failure Notification

Anytime the Seller detects an occurrence of a Failure Situation on a Product provided to the Buyer, the Seller will send a Failure Notification to the Buyer. The Incident details includes information about the nature of the Failure Situation and may include the expected closed date and time.

The following are additional requirements for the Failure Notification Use Case.

[R25] When sending the Failure Notification, the Seller **MUST** use `CimFailureIncident` type.

[R26] The Seller **MUST** set `CimFailureIncident.incidentType` attribute value to `failure`. [MEF173 R1]

[R27] The Seller **MUST** set `CimFailureIncident.impact` attribute value to `down`. [MEF173 R2]

[R28] The Seller **MUST** set `CimFailureIncident.description` attribute with sufficient details about the nature of the Failure Situation. [MEF173 R3]

[D7] When the `CimFailureIncident` is created, the Seller **SHOULD** set the `expectedClosedDate` to the estimated resolution date and time of the Failure Situation. [MEF173 D1]

## 6.4.2 Use case 2: Service Impairment Notification

Anytime the Seller detects an occurrence of a Service Impairment on a Product provided to the Buyer, the Seller will send a Service Impairment Notification to the Buyer. The Incident details includes information about the nature of the Service Impairment and may include the expected closed date and time.

The following are additional requirements for the Service Impairment Notification Use Case.

[R29] When sending the Failure Notification, the Seller **MUST** use `CimServiceImpairmentIncident` type.

[R30] If the Service Impairment is due to traffic violating traffic profiles associated with the subscribed service, such as traffic drops due to policing, the Seller **MUST** set `CimServiceImpairmentIncident.incidentType` attribute value to `policing`. [MEF173 R4]

[R31] If the Service Impairment is due to traffic congestion, such as resulting in packet drops or traffic delays, the Seller **MUST** set `CimServiceImpairmentIncident.incidentType` attribute value to `congestion`. [MEF173 R5]

[R32] The Seller **MUST** set `CimServiceImpairmentIncident.impact` attribute value to `degraded` or `intermittent`. [MEF173 R6]

[R33] The Seller **MUST** set `CimServiceImpairmentIncident.description` attribute with sufficient details about the nature of the Service Impairment. [MEF173 R7]

[D8] When the `CimServiceImpairmentIncident` is created, the Seller **SHOULD** set the `expectedClosedDate` to the estimated resolution date and time of the Service Impairment. [MEF173 D2]

A Buyer receiving multiple Service Impairment Notifications over a specific duration for a Product may use that as a trigger for negotiating changes to the Product-Specific Attributes (for example, Bandwidth Capacity, Bandwidth Profile, Class of Service, etc.).

## 6.4.3 Use case 3: Maintenance Notification

The Seller notifies the Buyer about an upcoming Planned Maintenance that could impact the operation of a Product provided to the Buyer. The Incident details includes the Planned Maintenance start and ending dates and times, to help the Buyer plan around the maintenance.

The following are additional requirements for the Planned Maintenance Notification Use Case.

**[R34]** When sending the Failure Notification, the Seller **MUST** use `CimMaintenanceIncident` type.

**[R35]** The Seller **MUST** set `CimMaintenanceIncident.incidentType` attribute value to `plannedMaintenance`. [MEF173 R8]

**[R36]** The Seller **MUST** set the `CimMaintenanceIncident.situationStartDate` attribute value to the Planned Maintenance start date and time. [MEF173 R9]

**[R37]** The Seller **MUST** set the `CimMaintenanceIncident.expectedClosedDate` to the Planned Maintenance ending date and time. [MEF173 R10]

The Seller Planned Maintenance notification should be sent days or weeks (depending on the Service Level Agreement) ahead of the activity.

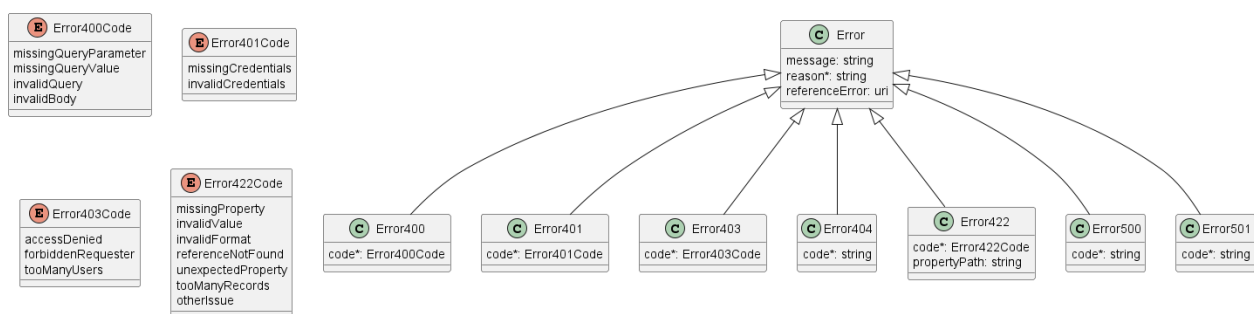
## 7. API Details

### 7.1. API patterns

#### 7.1.1. Indicating errors

Erroneous situations are indicated by appropriate HTTP responses. An error response is indicated by HTTP status 4xx (for client errors) or 5xx (for server errors) and appropriate response payload. The Product Order API uses the error responses as depicted and described below.

Implementations can use HTTP error codes not specified in this standard in compliance with rules defined in RFC 7231 [RFC7231]. In such a case, the error message body structure might be aligned with the `Error`.



**Figure 9: Data model types to represent an erroneous response**

##### 7.1.1.1. Type Error

**Description:** Standard Class used to describe API response error Not intended to be used directly. The `code` in the HTTP header is used as a discriminator for the type of error returned in runtime.

| Name    | Type   | Description                                                                                                      |
|---------|--------|------------------------------------------------------------------------------------------------------------------|
| message | string | Text that provides mode details and corrective actions related to the error. This can be shown to a client user. |

| Name           | Type                                     | Description                                                                      |
|----------------|------------------------------------------|----------------------------------------------------------------------------------|
| reason*        | string<br><small>maxLength = 255</small> | Text that explains the reason the for error. This can be shown to a client user. |
| referenceError | uri<br><small>format = uri</small>       | URL pointing to documentation describing the error                               |

### 7.1.1.2. Type Error400

**Description:** Bad Request. (<https://tools.ietf.org/html/rfc7231#section-6.5.1>)

Inherits from:

- [Error](#)

| Name  | Type                         | Description                                                                                                                                                                                                                                                                                             |
|-------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code* | <a href="#">Error400Code</a> | One of the following error codes: - missingQueryParameter: The URI is missing a required query-string parameter - missingQueryValue: The URI is missing a required query-string parameter value - invalidQuery: The query section of the URI is invalid. - invalidBody: The request has an invalid body |

### 7.1.1.3. **enum** Error400Code

**Description:** One of the following error codes:

- missingQueryParameter: The URI is missing a required query-string parameter
- missingQueryValue: The URI is missing a required query-string parameter value
- invalidQuery: The query section of the URI is invalid.
- invalidBody: The request has an invalid body

| Value                 | MEF 113                 |
|-----------------------|-------------------------|
| missingQueryParameter | MISSING_QUERY_PARAMETER |
| missingQueryValue     | MISSING_QUERY_VALUE     |
| invalidQuery          | INVALID_QUERY           |
| invalidBody           | INVALID_BODY            |

### 7.1.1.4. Type Error401

**Description:** Unauthorized. (<https://tools.ietf.org/html/rfc7235#section-3.1>)

Inherits from:

- [Error](#)

| Name  | Type                         | Description                                                                                                                                        |
|-------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| code* | <a href="#">Error401Code</a> | One of the following error codes: - missingCredentials: No credentials provided. - invalidCredentials: Provided credentials are invalid or expired |

### 7.1.1.5. **enum** Error401Code

**Description:** One of the following error codes:

- missingCredentials: No credentials provided.
- invalidCredentials: Provided credentials are invalid or expired

| Value              | MEF 113             |
|--------------------|---------------------|
| missingCredentials | MISSING_CREDENTIALS |
| invalidCredentials | INVALID_CREDENTIALS |

### 7.1.1.6. **Type** Error403

**Description:** Forbidden. This code indicates that the server understood the request but refuses to authorize it. (<https://tools.ietf.org/html/rfc7231#section-6.5.3>)

Inherits from:

- [Error](#)

| Name  | Type                         | Description                                                                                                                                                                                                                                |
|-------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code* | <a href="#">Error403Code</a> | This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes: - accessDenied: Access denied - forbiddenRequester: Forbidden requester - tooManyUsers: Too many users |

### 7.1.1.7. **enum** Error403Code

**Description:** This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes:

- accessDenied: Access denied
- forbiddenRequester: Forbidden requester
- tooManyUsers: Too many users

| Value              | MEF 113             |
|--------------------|---------------------|
| accessDenied       | ACCESS_DENIED       |
| forbiddenRequester | FORBIDDEN_REQUESTER |
| tooManyUsers       | TOO_MANY_USERS      |

### 7.1.1.8. **Type** Error404

**Description:** Resource for the requested path not found. (<https://tools.ietf.org/html/rfc7231#section-6.5.4>)

Inherits from:

- [Error](#)

| Name | Type | Description |
|------|------|-------------|
|------|------|-------------|

| Name | Type | Description |
|------|------|-------------|
|------|------|-------------|

|       |        |                                                                                                  |
|-------|--------|--------------------------------------------------------------------------------------------------|
| code* | string | The following error code: - notFound: A current representation for the target resource not found |
|-------|--------|--------------------------------------------------------------------------------------------------|

### 7.1.1.9. Type Error422

The response for HTTP status 422 is a list of elements that are structured using the Error422 data type. Each list item describes a business validation problem. This type introduces the `propertyPath` attribute which points to the erroneous property of the request, so that the Buyer may fix it easier. It is highly recommended that this property should be used, yet remains optional because it might be hard to implement.

**Description:** Unprocessable entity due to a business validation problem. (<https://tools.ietf.org/html/rfc4918#section-11.2>)

Inherits from:

- [Error](#)

| Name | Type | Description |
|------|------|-------------|
|------|------|-------------|

|       |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code* | <a href="#">Error422Code</a> | One of the following error codes: - missingProperty: The property the Seller has expected is not present in the payload - invalidValue: The property has an incorrect value - invalidFormat: The property value does not comply with the expected value format - referenceNotFound: The object referenced by the property cannot be identified in the Seller system - unexpectedProperty: Additional property, not expected by the Seller has been provided - tooManyRecords: the number of records to be provided in the response exceeds the Seller's threshold. - otherIssue: Other problem was identified (detailed information provided in a reason) |
|-------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|              |        |                                                                                                                                                                                                                                                                                                    |
|--------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| propertyPath | string | A pointer to a particular property of the payload that caused the validation issue. It is highly recommended that this property should be used. Defined using JavaScript Object Notation (JSON) Pointer ( <a href="https://tools.ietf.org/html/rfc6901">https://tools.ietf.org/html/rfc6901</a> ). |
|--------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.1.1.10. `enum` Error422Code

**Description:** One of the following error codes:

- missingProperty: The property the Seller has expected is not present in the payload
- invalidValue: The property has an incorrect value
- invalidFormat: The property value does not comply with the expected value format
- referenceNotFound: The object referenced by the property cannot be identified in the Seller system
- unexpectedProperty: Additional property, not expected by the Seller has been provided
- tooManyRecords: the number of records to be provided in the response exceeds the Seller's threshold.
- otherIssue: Other problem was identified (detailed information provided in a reason)

| Value | MEF 113 |
|-------|---------|
|-------|---------|

|                 |                  |
|-----------------|------------------|
| missingProperty | MISSING_PROPERTY |
|-----------------|------------------|

| Value              | MEF 113             |
|--------------------|---------------------|
| invalidValue       | INVALID_VALUE       |
| invalidFormat      | INVALID_FORMAT      |
| referenceNotFound  | REFERENCE_NOT_FOUND |
| unexpectedProperty | UNEXPECTED_PROPERTY |
| tooManyRecords     | TOO_MANY_RECORDS    |
| otherIssue         | OTHER_ISSUE         |

#### 7.1.1.11. Type Error500

**Description:** Internal Server Error. (<https://tools.ietf.org/html/rfc7231#section-6.6.1>)

Inherits from:

- [Error](#)

| Name | Type | Description |
|------|------|-------------|
|------|------|-------------|

|       |        |                                                                                                                                                                  |
|-------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code* | string | The following error code: - internalError: Internal server error - the server encountered an unexpected condition that prevented it from fulfilling the request. |
|-------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 7.1.1.12. Type Error501

**Description:** Not Implemented. (<https://tools.ietf.org/html/rfc7231#section-6.6.2>)

Inherits from:

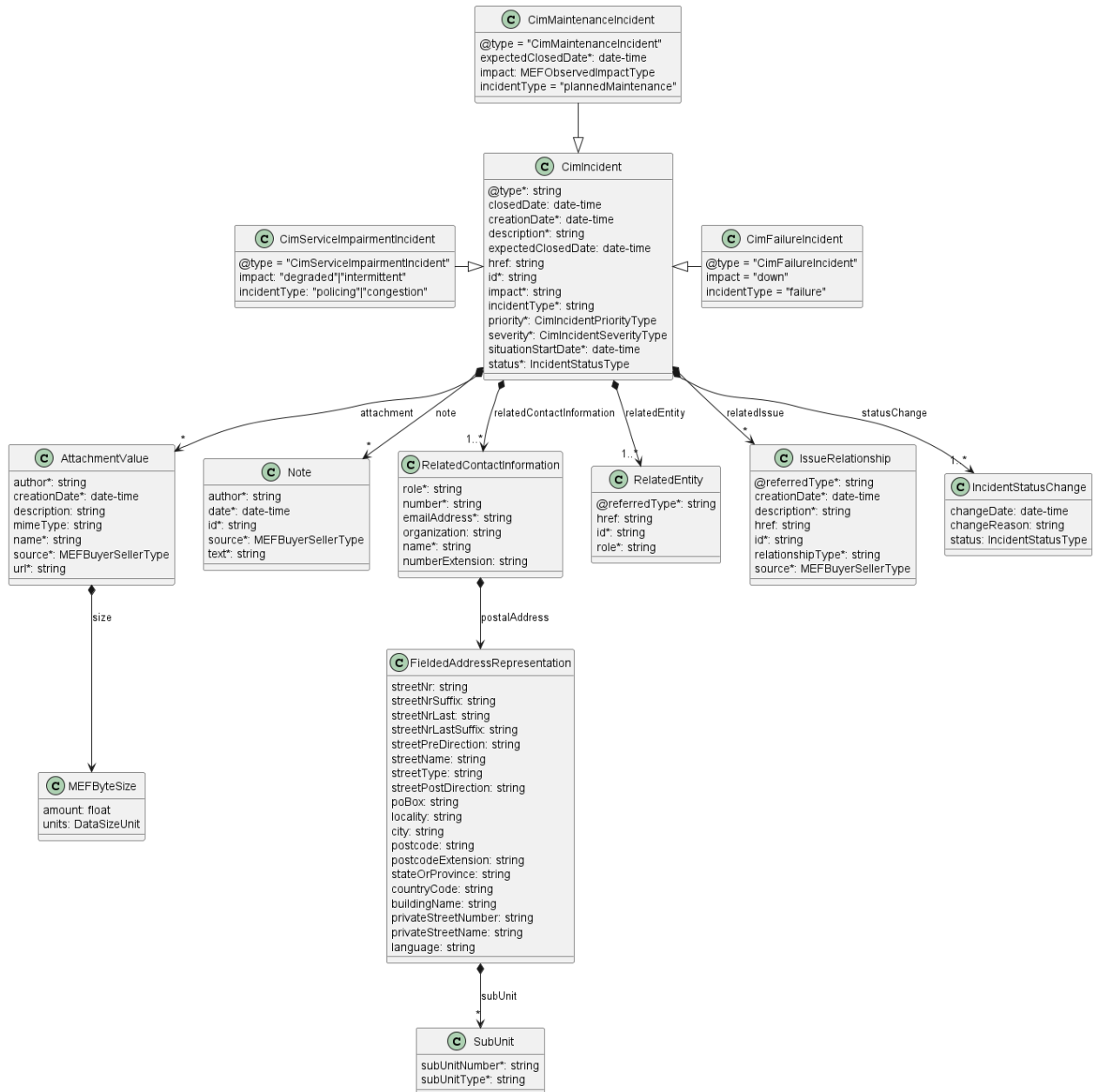
- [Error](#)

| Name | Type | Description |
|------|------|-------------|
|------|------|-------------|

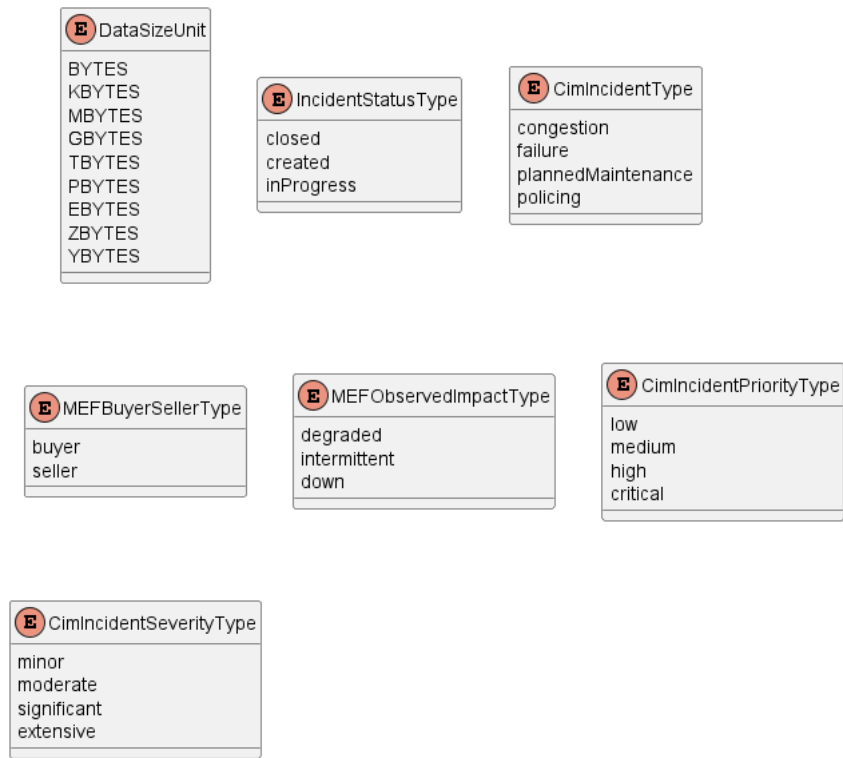
|       |        |                                                                                 |
|-------|--------|---------------------------------------------------------------------------------|
| code* | string | The following error code:: - notImplemented: Method not supported by the server |
|-------|--------|---------------------------------------------------------------------------------|

## 7.2. Management API Data model

Figure 10 presents the whole CIM data model followed by Figure 11 with all Enumeration data types. The rest of the section presents details of each Entity in tabular form



**Figure 10. CIM Management Data Model**



**Figure 11. CIM Management Data Model - Enums**

## 7.2.1. CimIncident

### 7.2.1.1. Type CimIncident

**Description:** An CIM Incident is a record of an issue that is not part of normal operation in the Seller's network that has a possible negative impact on the operability of the network on one or more Buyers.

| Name               | Type                                           | M/O | Description                                                                                                                                      | M<br>11 |
|--------------------|------------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| @type              | string                                         | O   | Used to unambiguously designate the class type when subclassing                                                                                  |         |
| attachment         | <a href="#">AttachmentValue[]</a>              | O   | Attachments to the Incident, such as a file or screenshot. Attachments may be added but may not be modified or deleted (for historical reasons). |         |
| closedDate         | date-time<br><small>format = date-time</small> | O   | The date the CIM Incident status was set to closed by the Seller                                                                                 |         |
| creationDate       | date-time<br><small>format = date-time</small> | O   | The date on which the CIM Incident was created                                                                                                   |         |
| description        | string                                         | O   | Description of the CIM Incident                                                                                                                  |         |
| expectedClosedDate | date-time<br><small>format = date-time</small> | O   | The date provided by the Seller to indicate when the CIM Incident is expected to be closed.                                                      |         |
| href               | string                                         | O   | Hyperlink, a reference to the CIM Incident entity                                                                                                |         |

| Name                      | Type                                                                       | M/O | Description                                                                                                                                                                                                                                                |
|---------------------------|----------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| id                        | string                                                                     | O   | Unique (within the Seller domain) identifier for the CIM Incident.                                                                                                                                                                                         |
| impact                    | string                                                                     | O   | The presumed impact on the Buyer for the referenced Product(s).                                                                                                                                                                                            |
| incidentType              | string                                                                     | O   | The presumed cause of the CIM Incident as evaluated by the Seller.                                                                                                                                                                                         |
| note                      | <a href="#">Note[]</a>                                                     | O   | A set of unstructured comments or information associated to the CIM Incident. Notes may be added but may not be modified or deleted (for historical reasons).                                                                                              |
| priority                  | <a href="#">CimIncidentPriorityType</a>                                    | O   | The priority (ITIL) is based on the assessment of the impact and urgency of how quickly the CIM Incident should be resolved after evaluation by the Seller of the impact of the CIM Incident.                                                              |
| relatedContactInformation | <a href="#">RelatedContactInformation[]</a><br><small>minItems = 1</small> | O   | Party playing a role in this CIM Incident. The 'role' is to specify the type of contact as specified in MEF 113: Incident Contact ('role=incidentContact') - REQUIRED to be set by the Seller Incident Technical Contact ('role=incidentTechnicalContact') |
| relatedEntity             | <a href="#">RelatedEntity[]</a><br><small>minItems = 1</small>             | O   | A set of identifiers of the Products on which the CIM Incident could have an impact on the normal operation.                                                                                                                                               |
| relatedIssue              | <a href="#">IssueRelationship[]</a>                                        | O   | A list of Related Issue relationships. Represents relationships to other Trouble Tickets, Incidents, and CIM Incidents.                                                                                                                                    |
| severity                  | <a href="#">CimIncidentSeverityType</a>                                    | O   | The severity (ITIL) of the CIM Incident as evaluated by the Seller.                                                                                                                                                                                        |
| situationStartDate        | date-time<br><small>format = date-time</small>                             | O   | The date when the situation was first identified, for example via error logs.                                                                                                                                                                              |
| status                    | <a href="#">IncidentStatusType</a>                                         | O   | The current status of the CIM Incident                                                                                                                                                                                                                     |

| Name         | Type                                                                  | M/O | Description                                                                                | M<br>11 |
|--------------|-----------------------------------------------------------------------|-----|--------------------------------------------------------------------------------------------|---------|
| statusChange | <a href="#">IncidentStatusChange[]</a><br><small>minItems = 1</small> | O   | The status change history that is associated to the CIM Incident. Populated by the Seller. |         |

### 7.2.1.2. Type CimFailureIncident

**Description:** Sent when the Seller detects an occurrence of a Situation on a Product provided to the Buyer, where the Product is non-operational (for example, down, routing failures, severe packet drops, etc.).

Inherits from:

- [CimIncident](#)

| Name         | Type   | M/O | Description                                                        | MEF<br>113 |
|--------------|--------|-----|--------------------------------------------------------------------|------------|
| @type        | string | O   | Used to unambiguously designate the class type when using `oneOf`  |            |
| impact       | string | O   | The presumed impact on the Buyer for the referenced Product(s).    |            |
| incidentType | string | O   | The presumed cause of the CIM Incident as evaluated by the Seller. |            |

### 7.2.1.3. Type CimMaintenanceIncident

**Description:** Any scheduled maintenance to be performed by the Seller that may impact a Product for the Buyer, which also includes the planned ending time and date.

Inherits from:

- [CimIncident](#)

| Name         | Type                                  | M/O | Description                                                        | MEF<br>113 |
|--------------|---------------------------------------|-----|--------------------------------------------------------------------|------------|
| @type        | string                                | O   | Used to unambiguously designate the class type when using `oneOf`  |            |
| impact       | <a href="#">MEFObservedImpactType</a> | O   | The presumed impact on the Buyer for the referenced Product(s).    |            |
| incidentType | string                                | O   | The presumed cause of the CIM Incident as evaluated by the Seller. |            |

### 7.2.1.4. Type CimServiceImpairmentIncident

**Description:** Sent when the Seller detects an occurrence of a Situation on a Product provided to the Buyer, where the Product is not meeting the Product specifications or is not operational on an intermittent basis. This includes impairments due to traffic violating traffic profile thresholds (for example, excessive traffic drops due to policing) and/or traffic congestion exceeding thresholds.

Inherits from:

- [CimIncident](#)

| Name         | Type   | M/O | Description                                                        | MEF 113 |
|--------------|--------|-----|--------------------------------------------------------------------|---------|
| @type        | string | O   | Used to unambiguously designate the class type when using `oneOf`  |         |
| impact       | string | O   | The presumed impact on the Buyer for the referenced Product(s).    |         |
| incidentType | string | O   | The presumed cause of the CIM Incident as evaluated by the Seller. |         |

#### 7.2.1.5. **enum** CimIncidentPriorityType

**Description:** Possible values for the priority of CIM Incident

| Value    | MEF 113  |
|----------|----------|
| low      | LOW      |
| medium   | MEDIUM   |
| high     | HIGH     |
| critical | CRITICAL |

#### 7.2.1.6. **enum** CimIncidentSeverityType

**Description:** Possible values for the severity of CIM Incident

| Value       | MEF 113     |
|-------------|-------------|
| minor       | MINOR       |
| moderate    | MODERATE    |
| significant | SIGNIFICANT |
| extensive   | EXTENSIVE   |

#### 7.2.1.7. **enum** CimIncidentType

**Description:** Possible values for the type of the CIM Incident:

- congestion - Any Situation due to traffic congestion, such as resulting in packet drops or traffic delays.
- failure - Any Situation where the Product is non-operational, such as down, routing fail-ures or severe packet drops.
- plannedMaintenance -Any scheduled maintenance that may impact a Product for the Buyer.
- policing - Any Situation due to traffic violating traffic profiles as-sociated with the subscribed service, such as traffic drops due to policing.

| Value | MEF 113 |
|-------|---------|
|-------|---------|

| Value              | MEF 113             |
|--------------------|---------------------|
| congestion         | CONGESTION          |
| failure            | FAILURE             |
| plannedMaintenance | PLANNED_MAINTENANCE |
| policing           | POLICING            |

#### 7.2.1.8. Type IncidentStatusChange

**Description:** Holds the status notification reasons and associated date the status changed, populated by the server

| Name         | Type                                           | M/O | Description                           | MEF 113                    |
|--------------|------------------------------------------------|-----|---------------------------------------|----------------------------|
| changeDate   | date-time<br><small>format = date-time</small> | O   | The date and time the status changed. | Not represented in MEF 113 |
| changeReason | string                                         | O   | The reason why the status changed.    | Not represented in MEF 113 |
| status       | IncidentStatusType                             | O   | Reached status                        | Not represented in MEF 113 |

#### 7.2.1.9. enum IncidentStatusType

**Description:** Possible values for the status of the Incident

| status     | MEF 113 name | Description                                                                                                                                                               |
|------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| closed     | CLOSED       | The Situation described in the Incident was closed by has been resolved and normal operation has been restored on the Seller.Seller's network. This is a terminal status. |
| created    | CREATED      | A new Incident has been created and allocated a unique <b>id</b> .                                                                                                        |
| inProgress | IN_PROGRESS  | The Incident is in the process of being handled and investigated for resolution by the Seller.                                                                            |
| Value      | MEF 113      |                                                                                                                                                                           |
| closed     | CLOSED       |                                                                                                                                                                           |
| created    | CREATED      |                                                                                                                                                                           |
| inProgress | IN_PROGRESS  |                                                                                                                                                                           |

#### 7.2.1.10. Type IssueRelationship

**Description:** Represents relationships to other Trouble Tickets and Incidents

| Name          | Type                                           | M/O | Description                                                | MEF 113                |
|---------------|------------------------------------------------|-----|------------------------------------------------------------|------------------------|
| @referredType | string                                         | M   | The type of the referred Issue (Incident or TroubleTicket) | Related Object Type    |
| creationDate  | date-time<br><small>format = date-time</small> | M   | The date the relationship was created                      | Relation Creation Date |

| Name             | Type                               | M/O | Description                                                                                   | MEF 113                     |
|------------------|------------------------------------|-----|-----------------------------------------------------------------------------------------------|-----------------------------|
| description      | string                             | M   | A description of the reason for the Relation Source to set the relationship                   | Relation Reason Description |
| href             | string                             | O   | Reference of the Trouble Ticket or Incident                                                   | Not represented in MEF 113  |
| id               | string                             | M   | Unique identifier of the referenced Issue (Trouble Ticket or Incident)                        | Related Object Identifier   |
| relationshipType | string                             | M   | Type of the Trouble Ticket relationship can be blocks, depends on, duplicates, causes, etc... | Relation Type               |
| source           | <a href="#">MEFBuyerSellerType</a> | M   | Indicates if this Related Issue was added by the Buyer or the Seller.                         | Relation Source             |

#### 7.2.1.11. **enum** MEFObservedImpactType

**Description:** An enumeration of the possible values of impact observed by the Buyer.

- degraded: When the Product is impacted and not meeting the Product specifications.
- intermittent: When the Product is not operational as intended on an intermittent basis.
- down: When the Product is non-operational.

| Value        | MEF 113      |
|--------------|--------------|
| degraded     | DEGRADED     |
| intermittent | INTERMITTENT |
| down         | DOWN         |

### 7.2.2. Common

Types described in this subsection are shared among two or more Cantata and Sonata APIs.

#### 7.2.2.1. Type AttachmentValue

**Description:** Complements the description of an element (for instance a product) through video, pictures...

| Name         | Type                                           | M/O | Description                                                      | MEF 113           |
|--------------|------------------------------------------------|-----|------------------------------------------------------------------|-------------------|
| author       | string                                         | M   | The name of the person or organization who added the Attachment. | Attachment Author |
| creationDate | date-time<br><small>format = date-time</small> | M   | The date the Attachment was added.                               | Attachment Date   |
| description  | string                                         | O   | A narrative text describing the content of the attachment        | Description       |

| Name     | Type               | M/O | Description                                                                 | MEF 113           |
|----------|--------------------|-----|-----------------------------------------------------------------------------|-------------------|
| mimeType | string             | O   | Attachment mime type such as extension file for video, picture and document | Mime Type         |
| name     | string             | M   | The name of the attachment                                                  | Attachment Name   |
| size     | MEFByteSize        | O   | The size of the attachment.                                                 | Size              |
| source   | MEFBuyerSellerType | M   | Indicates if the attachment was added by the Buyer or the Seller.           | Attachment Source |
| url      | string             | M   | URL where the attachment is located.                                        | URL               |

#### 7.2.2.2. **enum DataSizeUnit**

**Description:** The unit of measure in the data size.

| Value  | MEF 113 |
|--------|---------|
| BYTES  | BYTES   |
| KBYTES | KBYTES  |
| MBYTES | MBYTES  |
| GBYTES | GBYTES  |
| TBYTES | TBYTES  |
| PBYTES | PBYTES  |
| EBYTES | EBYTES  |
| ZBYTES | ZBYTES  |
| YBYTES | YBYTES  |

#### 7.2.2.3. **Type FieldedAddressRepresentation**

**Description:** A type of Address that has a discrete field and value for each type of boundary or identifier down to the lowest level of detail. For example "street number" is one field, "street name" is another field, etc.

| Name               | Type   | M/O | Description                                                                                                           | MEF 113 |
|--------------------|--------|-----|-----------------------------------------------------------------------------------------------------------------------|---------|
| streetNr           | string | O   | Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses. |         |
| streetNrSuffix     | string | O   | The first street number suffix (in a street number range) or the suffix for the street number if there is no range    |         |
| streetNrLast       | string | O   | Last number in a range of street numbers allocated to an Address                                                      |         |
| streetNrLastSuffix | string | O   | Last street number suffix for a ranged Address                                                                        |         |

| Name                | Type                                                     | M/O | Description                                                                                                                         | MEF<br>113 |
|---------------------|----------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------|------------|
| streetPreDirection  | string                                                   | O   | The direction of the street that appears before the Street Name                                                                     |            |
| streetName          | string                                                   | O   | Name of the street or other street type                                                                                             |            |
| streetType          | string                                                   | O   | The type of street (e.g., alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf) |            |
| streetPostDirection | string                                                   | O   | A modifier denoting a relative direction that appears after the Street Name.                                                        |            |
| poBox               | string                                                   | O   | Number identifying a specific location in a post office.                                                                            |            |
| locality            | string                                                   | O   | An area of defined or undefined boundaries within a local authority or other legislatively defined area.                            |            |
| city                | string                                                   | O   | City in which the Address is located.                                                                                               |            |
| postcode            | string                                                   | O   | A descriptor for a postal delivery area used to speed and simplify the delivery of mail (also known as zip code)                    |            |
| postcodeExtension   | string                                                   | O   | The extension used on a postal code. Note: there are different use codes for this attribute depending upon the country.             |            |
| stateOrProvince     | string                                                   | O   | The State or Province in which the Address is located.                                                                              |            |
| countryCode         | string<br><small>minLength = 2<br/>maxLength = 2</small> | O   | Country in which the Address is located, defined using two characters as defined in ISO 3166                                        |            |
| subUnit             | SubUnit[]                                                | O   | The Sub Unit represented as a list. This is a list to allow complex sub-unit information such as SUITE 42 ROOM A                    |            |
| buildingName        | string                                                   | O   | The well-known name of a building that is located at this Address (e.g., where there is one Address for a campus).                  |            |
| privateStreetNumber | string                                                   | O   | Street number on a private street within the Address.                                                                               |            |
| privateStreetName   | string                                                   | O   | Private streets internal to a property (e.g., a university) may have internal names that are not recorded by the land title office. |            |
| language            | string<br><small>minLength = 2<br/>maxLength = 2</small> | O   | The language in which the address is expressed. It MUST use the ISO 639:2023 two letter code 639:2023                               |            |

#### 7.2.2.4. enum MEFBuyerSellerType

**Description:** An enumeration with buyer and seller values.

**Value MEF 113**

buyer BUYER

seller SELLER

**7.2.2.5. Type MEFByteSize****Description:** A size represented by value and Byte units

| Name   | Type                                                   | M/O | Description                   | MEF 113 |
|--------|--------------------------------------------------------|-----|-------------------------------|---------|
| amount | float<br><small>default = 1<br/>format = float</small> | O   | Numeric value in a given unit | Value   |
| units  | DataSizeUnit                                           | O   | Byte Unit                     | Unit    |

**7.2.2.6. Type Note****Description:** Extra information about a given entity. Only useful in processes involving human interaction. Not applicable for an automated process.

| Name   | Type                                           | M/O | Description                                                                                                                   | MEF 113                    |
|--------|------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| author | string                                         | M   | Author of the note                                                                                                            | Note<br>Author             |
| date   | date-time<br><small>format = date-time</small> | M   | Date of the note                                                                                                              | Note Date                  |
| id     | string                                         | M   | Identifier of the note within its containing entity (may or may not be globally unique, depending on provider implementation) | Not represented in MEF 113 |
| source | MEFBuyerSellerType                             | M   | Indicates if this Note was added by the Buyer or Seller.                                                                      | Note<br>Source             |
| text   | string                                         | M   | Text of the note                                                                                                              | Note Text                  |

**7.2.2.7. Type RelatedContactInformation****Description:** Contact data for a person or organization that is involved in the product offering qualification. In a given context it is always specified by the Seller (e.g. Seller Contact Information) or by the Buyer.

| Name   | Type   | M/O | Description                                     | MEF 113                     |
|--------|--------|-----|-------------------------------------------------|-----------------------------|
| role   | string | M   | A role of the particular contact in the request | Not represented in MEF 113  |
| number | string | M   | Phone number                                    | Contract<br>Phone<br>Number |

| Name            | Type                         | M/O | Description                                                            | MEF 113                         |
|-----------------|------------------------------|-----|------------------------------------------------------------------------|---------------------------------|
| emailAddress    | string                       | M   | Email address                                                          | Contact email Address           |
| postalAddress   | FieldedAddressRepresentation | O   | Identifies the postal address of the person or office to be contacted. | Contact Postal Address          |
| organization    | string                       | O   | The organization or company that the contact belongs to                | Contact Organization            |
| name            | string                       | M   | Name of the contact                                                    | Contact Name                    |
| numberExtension | string                       | O   | Phone number extension                                                 | Contract Phone Number Extension |

#### 7.2.2.8. Type RelatedEntity

**Description:** A reference to an entity, where the type of the entity is not known in advance.

| Name          | Type                                       | M/O | Description                                                            | MEF 113                    |
|---------------|--------------------------------------------|-----|------------------------------------------------------------------------|----------------------------|
| @referredType | string<br><small>default = Product</small> | M   | The actual type of the target instance when needed for disambiguation. | Not represented in MEF 113 |
| href          | string                                     | O   | Reference of the related entity.                                       | Not represented in MEF 113 |
| id            | string                                     | M   | Unique identifier of a related entity.                                 | Product Identifier         |
| role          | string                                     | M   | The role of an entity.                                                 | Not represented in MEF 113 |

#### 7.2.2.9. Type SubUnit

**Description:** Allows for sub unit identification

| Name          | Type   | M/O | Description                                                                                 | MEF 113 |
|---------------|--------|-----|---------------------------------------------------------------------------------------------|---------|
| subUnitNumber | string | M   | The discriminator used for the subunit, often just a simple number but may also be a range. |         |
| subUnitType   | string | M   | The type of subunit e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.                |         |

### 7.2.3. Notification registration

Notification registration and management are done through [/hub](#) API endpoint. The below sections describe data models related to this endpoint.

### 7.2.3.1. Type EventSubscriptionInput

**Description:** This class is used to register for Notifications.

| API name                                              | MEF 113 name          | Description                                              |
|-------------------------------------------------------|-----------------------|----------------------------------------------------------|
| <a href="#">cimIncidentCreateEvent</a>                | INCIDENT_CREATE       | A new CIM Incident was created by the Seller.            |
| <a href="#">cimIncidentAttributeValueChangedEvent</a> | INCIDENT_UPDATE       | An open CIM Incident was updated by the Seller.          |
| <a href="#">cimIncidentStatusChangeEvent</a>          | INCIDENT_STATE_CHANGE | An CIM Incident <b>status</b> was changed by the Seller. |

| Name     | Type   | M/O | Description                                                                                                                                                                                                                                                                                                                |
|----------|--------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| callback | string | M   | This callback value must be set to *host* property from Buyer Notification. This property is appended with the base path and notification resource path : notification is sent. E.g. for "callback": "http://buyer.mef.com/listenerEndpoint" `http://buyer.mef.com/listenerEndpoint/mefApi/sonata/cimIncidentNotification` |
| query    | string | O   | This attribute is used to define to which type of events to subscribe for more than one event type. To subscribe for more than one event type use `eventType=cimIncidentAttributeValueChangedEvent,cimIncidentCreateEvent`. ending in subscription for all event types.                                                    |

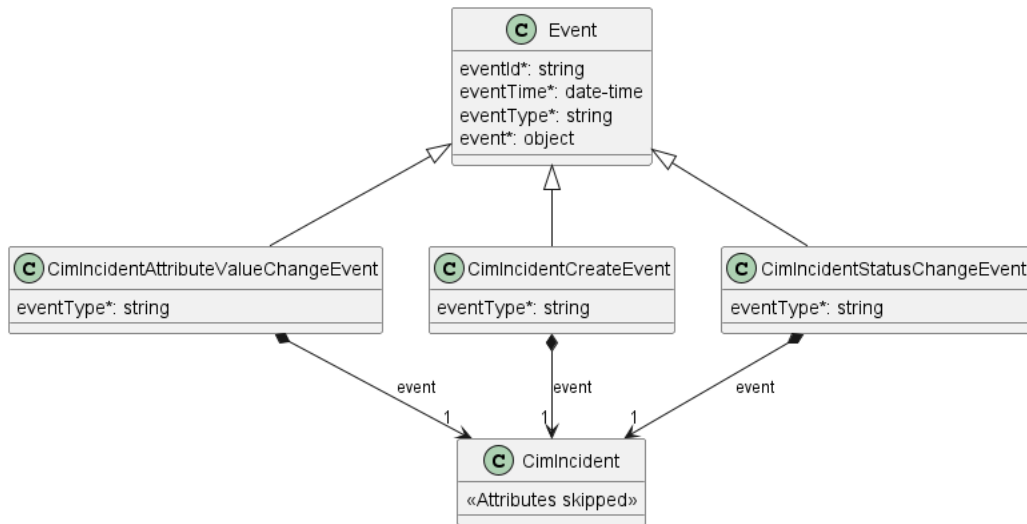
### 7.2.3.2. Type EventSubscription

**Description:** Sets the communication endpoint address the service instance must use to deliver notification information

| Name     | Type   | M/O | Description                                                                                  | MEF 113                                  |
|----------|--------|-----|----------------------------------------------------------------------------------------------|------------------------------------------|
| callback | string | M   | The value provided by the Buyer in 'EventSubscriptionInput' during notification registration | Notification Target Information          |
| id       | string | M   | An identifier of the event subscription assigned by the Seller when a resource is created.   | Not represented in MEF 113               |
| query    | string | O   | This attribute is used to define notification registration constraints.                      | List of Event Notification Types, Action |

## 7.3. Notification API Data model

Figure 12 presents the CIM Notification data model.



**Figure 12. CIM Notification Data Model**

This data model is used to construct requests and responses of the API endpoints described in [Section 5.2.2](#).

### 7.3.1. Type Event

**Description:** Event class is used to describe information structure used for notification.

| Name      | Type                                           | M/O | Description                                      | MEF 113 |
|-----------|------------------------------------------------|-----|--------------------------------------------------|---------|
| eventId   | string                                         | M   | Id of the event                                  |         |
| eventTime | date-time<br><small>format = date-time</small> | M   | Date-time when the event occurred                |         |
| eventType | string                                         | M   | The type of the notification.                    |         |
| event     | object                                         | M   | The event linked to the involved resource object |         |

### 7.3.2. Type CimIncidentAttributeValueChangedEvent

**Description:**

Inherits from:

- [Event](#)

| Name      | Type                        | M/O | Description                                                       | MEF 113 |
|-----------|-----------------------------|-----|-------------------------------------------------------------------|---------|
| eventType | string                      | M   | Indicates the type of the event.                                  |         |
| event     | <a href="#">CimIncident</a> | M   | A reference to the object that is the source of the notification. |         |

### 7.3.3. Type CimIncidentCreateEvent

**Description:**

Inherits from:

- [Event](#)

| Name      | Type                        | M/O | Description                                                       | MEF<br>113 |
|-----------|-----------------------------|-----|-------------------------------------------------------------------|------------|
| eventType | string                      | M   | Indicates the type of the event.                                  |            |
| event     | <a href="#">CimIncident</a> | M   | A reference to the object that is the source of the notification. |            |

#### 7.3.4. Type CimIncidentStatusChangeEvent

##### Description:

Inherits from:

- [Event](#)

| Name      | Type                        | M/O | Description                                                       | MEF<br>113 |
|-----------|-----------------------------|-----|-------------------------------------------------------------------|------------|
| eventType | string                      | M   | Indicates the type of the event.                                  |            |
| event     | <a href="#">CimIncident</a> | M   | A reference to the object that is the source of the notification. |            |

## 8. References

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